

CHANDANA ANAND RANGAPPA

California(Open to relocate) | +1(408)-426 6291 | canan003@ucr.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Full-stack developer and graduate researcher experienced in building scalable web applications using frameworks like React, Flask. Proven skills in Python, C++, Java, SQL, with a focus on high-performance, reliable software development. Demonstrated ability to collaborate effectively with cross-functional teams in agile environments, ensuring streamlined cloud deployment and robust API integrations.

EDUCATION

UNIVERSITY OF CALIFORNIA, RIVERSIDE, Riverside, CA, USA

Sep 2024 - Mar 2026

Master of Science, Computer Engineering

- **Coursework:** Artificial Intelligence, Big Data Analytics, Compiler Construction, Machine Learning, Spatial Computing, Distributed Systems, Advanced Operating Systems, Data Mining, Information Retrieval, DBMS

DAYANANDA SAGAR COLLEGE OF ENGINEERING, Bengaluru, Karnataka, India

Sep 2020 - Jun 2024

Bachelor of Engineering, Computer Science & Engineering

- **Coursework:** Data Structures, Machine Learning, Big-Data, DBMS, Software Engineering, AI, Cloud Computing, Web Development, Cyber Security, Object-Oriented Programming in Java, Discrete Mathematics, Algorithms, Data Science, Human Computer Interface, Natural Language Processing, IoT

WORK EXPERIENCE

Anand Diagnostic Laboratory, *FULL-STACK DEVELOPMENT INTERN*, Bengaluru, India

Dec 2023 - May 2024

- Developed a scalable Antibiotic Susceptibility Testing system using YOLOv5, OpenCV, TensorFlow, and REST API for ZOI detection, aligning with modern web application standards.
- Improved throughput and data quality by achieving 95% accuracy and 30% faster reporting through ML automation, demonstrating effective problem-solving and collaboration with cross-functional teams.
- Engineered and deployed a full-stack application using React and Flask with RESTful APIs for real-time uploads, testing, and results visualization, adhering to front-end and back-end development best practices.
- Produced Docker images and deployed inference services on AWS EC2, ensuring scalable, reliable processing and consistent integration with cloud-based solutions.
- Implemented automated pipelines for inference, utilized MySQL for data storage, and incorporated data validation using NumPy, highlighting a commitment to agile development and software engineering principles.

University of California, Riverside, *GRADUATE RESEARCHER*, Riverside, CA

Sep 2025 - Oct 2025

- Analyzed and detected 'missing-handler' Promise bugs in JavaScript/Node.js using static analysis tools (NullAway) to enhance code quality and API integration reliability.
- Validated rules and experimental results, refining precision and coverage beyond ESLint no-floating-promises, thereby supporting robust software testing and agile methodologies.

PROJECTS

.edu Web Crawler - [GitHub](#)

- Built a Python BFS crawler indexing 500MB across 50K .edu domains with SHA-256 deduplication for robust reliability.
- Designed modular, batch storage to speed ingestion and improve downstream search/indexing retrievability at scale.

Operating Systems Lab - XV6 Programs Implementation - [GitHub](#)

- Implemented system calls and Lottery and Stride scheduling in xv6 to improve CPU fairness in benchmarks.
- Built a user-level threading library with clone and join in C, demonstrating low-level concurrency and scheduling trade-offs.

U-ASK: Spatial-Textual Search Engine - [GitHub](#)

- Developed a spatial-textual query engine with PostgreSQL & OpenSearch to support location-aware keyword search.
- Implemented POWER and Boolean Range ranking algorithms for efficient top-k retrieval and negative keyword filtering.

Compiler Construction - [GitHub](#)

- Built LLVM passes for constant propagation & SSA-based optimizations in C++, improving execution efficiency by 30%.
- Enhanced debugging and code optimization workflows, demonstrating strong systems-level engineering skills.

SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, TypeScript, SQL, HTML, CSS, Go, C#
- **Frameworks and Tools:** React.js, Node.js, Flask, PyTorch, TensorFlow, YOLOv5, OpenCV, Pandas, NumPy, Matplotlib, .NET
- **Systems:** Linux, Concurrency, Memory management, Profiling, Debugging, Systems Monitoring
- **Cloud & DevOps:** AWS (ECS, Lambda, SQS), EC2, Docker, Kubernetes, CI/CD pipelines, Git, Agile/Scrum
- **Databases & Big Data:** MySQL, PostgreSQL, MongoDB, Hadoop, HDFS, Spark, DynamoDB, Data Modelling, LLMs
- **Software Engineering:** RESTful API development, Unit & Automation Testing, Software Development Life Cycle

PUBLICATIONS

Inhibition Zones: A Window into Antibiotic Resistance - [Paper](#) | IoT Solutions for Food Quality Monitoring - [Paper](#)

ACHIEVEMENTS

Student Services Fee Advisory Committee (Subcommittee Chair) | University of California, Riverside